MANDATORY PROCESS SEQUENCE

EVERY STEP IN ORDER — NO SUBSTITUTION

1 — MACHINE

VC-101 base + VC-102 lid. Posts integral. Fill port drilled + counterbored.

2 — CLEAN — no hydrocarbons

Vapour degrease → 10% H2SO4 pickle 60 s → DI rinse → N2 blow dry.

3 — SINTER WICK — per VC-103

Cu powder 45-75 um. 950 C / 30 min / H2 or vacuum. Mask rim, post tops, fill bore.

4 — DIFFUSION BOND — Cu-Cu solid state

Stack lid on base. 850 C / 8 MPa / 60 min / vacuum < 1e-3 Pa.

5 — WELD FILL TUBE — after bonding

Laser or micro-TIG weld Cu tube OD3.0 to lid. No filler, no flux.

6 — LEAK TEST 1 — gross + fine

He mass spec, chamber under vacuum. Reject > 1e-9 Pa.m3/s.

7 — BAKE-OUT — removes adsorbed gas

200 C / 4 h / < 1e-3 Pa through the fill tube. Pump throughout.

8 — CHARGE — ± 0.05 mL

1.90 mL DI water 18 MOhm.cm, deaerated. Burette under vacuum, chamber at 0 C.

9 — DEGAS — removes dissolved O2

Freeze-pump-thaw x 3. Final pump at ice point to < 10 Pa.

10 — PINCH-OFF — NO screw seal

Cold-weld the tube in a hydraulic pincher. Laser-seal the tip. Trim + fold.

11 — LEAK TEST 2 — post-seal

He bomb 3 bar / 4 h, then mass spec. Reject > 1e-9 Pa.m3/s.

12 — BURN-IN — stabilises wick

3 thermal cycles 20 ↔ 100 C. Re-test R th after.

13 — ACCEPT — final gate

R th (case-to-case) < 0.25 K/W at 50 W, horizontal and vertical.

PROCESS PARAMETERS

PARAMETER	VALUE	LIMIT
Working fluid	**DI water 18 MOhm.cm**	deaerated
Charge volume	**1.90 mL**	± 0.05 mL
Wick void volume	1.51 mL	charge = 125 %
Bake-out	200 C / 4 h	< 1e-3 Pa
Final vacuum before seal	< 10 Pa	at ice point
Diffusion bond	850 C / 8 MPa / 60 min	vacuum
Leak rate	**< 1e-9 Pa.m3/s**	He mass spec
Acceptance R th	**< 0.25 K/W @ 50 W**	both orientations

PROHIBITED — CAUSES FIELD FAILURE

- Charging before diffusion bonding. Water boils at 100 C and the bond runs at 850 C: the chamber ruptures. This error appears on the superseded Maxsorb drawing — do not repeat it.
- Threaded or O-ring fill seals. A screw seal cannot hold high vacuum; helium ingress accumulates as non-condensable gas and the condenser is blanketed within months. Cold-weld pinch-off only.
- Aluminium anywhere wetted. Al + water generates H2. The fin stack is external and TIM-coupled only.
- Undeaerated water. Dissolved O2 oxidises copper and liberates H2.
- Tough-pitch copper (C11000 / JIS C1100) or brass/bronze anywhere in the envelope. Oxygen-bearing copper cracks in the hydrogen sinter; Zn and Sn evaporate under vacuum at bond temperature and contaminate the wick.
- Activated carbon or any organic inside the envelope without a 250 C / 8 h vacuum bake.

Non-condensable gas is the dominant failure mode of a sealed two-phase device.

Steps 7, 9 and 10 exist solely to prevent it. None of them is optional.

QUALITY RECORDS

STEP	RECORD RETAINED	HOLD POINT
4	bond furnace chart (T, P, vacuum)	yes — first article
6	He leak rate, pre-charge	**yes — 100 %**
7	bake-out pressure trace	yes
8	charge mass, by weight	**yes — 100 %**
11	He leak rate, post-seal	**yes — 100 %**
13	R th, both orientations	**yes — 100 %**

Charge by MASS, not by volume: 1.90 mL DI water = 1.87 g at 20 C. A burette

reading is not traceable once the chamber is cold and under vacuum.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK
TITLE: **PROCESS & CHARGING SPECIFICATION** — Cu-H2O VAPOUR CHAMBER / MANDATORY SEQUENCE			SCALE: **NONE** — THIRD ANGLE PROJECTION	UNITS: **mm**
DRAWING No.: **VC-300**			REV: **A**	SHEET: **6 / 7**
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-09